Analyzing historically-disputed events in online encyclopedias

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1 Related Work

1.1 Editorial Conflict and Structural Controversy Detection

Early studies of controversy on Wikipedia focused on how disagreements appear through the way users edit and interact with articles. Yasseri et al. (2012) formalized editorial conflict via the mutual reversion metric, showing that disputes manifest as consistent reverting of edits and long-lasting disagreements on talk pages. Yasseri et al. (2014) expanded on this crosslingually to reveal that while some controversial topics are shared globally, most are language dependent. Moving to controversy at the topic-level, Borra et al. (2015) introduced *Contropedia* to track controversy measured at specific wikilinks and sections, while (Bykau et al., 2015) introduced the Controversy Detection Algorithm (CDA), which tracks repeated changes in text and hyperlinks across revision histories to identify specific controversies within an article. Benslimane et al. (2021) combined user interaction graph neural networks with text representations to identify controversy in online discussions such as Reddit, while mentioning that it could be expanded to online encyclopedias.

1.2 Cross-Lingual Framing and Collective Memory Bias

Beyond metadata, independent language editions diverge significantly in narrating inter-group conflict. Oeberst et al. (2020) found that communities consistently describe their own group as more moral and less culpable. Analyzing 2,542 battles in 20 languages, Xia and Gomez-Zara (2026) used connotation frames to measure power, agency, and sentiment, and found significant differences in the narrative when language communities were involved, while uninvolved communities tended to be more similar. Creanga et al. (2026) study three contested events in Romanian history across five editions (Romanian, Hungarian, Russian, Turkish, and English). They report "citation isolation", with only 2 of 119 citations shared across three editions of the Battle of Posada article, and track LLM-assigned narrative scores across article snapshots from 2005 to 2024. At the sentence level, Aleksandrova and Lareau (2019) developed multilingual classifiers to detect Neutral Point of View (NPOV) violations, while Borkakoty and Espinosa-Anke (2024) benchmarked automated methods for distinguishing hoaxes from legitimate encyclopedic articles. He et al. (2018) showed that Wikipedia language editions often use substantially different images to represent the same concepts, which may partly reflect cultural contextualization.

1.3 Geopolitical Bias in LLMs and AI Encyclopedias

As large language models (LLMs) ingest encyclopedic text, cross-lingual divergences propagate into generative outputs. Li et al. (2024a) curated *BorderLines*, showing that multilingual LLMs systematically align territorial sovereignty answers with the prompt language's geopolitical stance. In retrieval-augmented generation (RAG), Li et al. (2024b) found that multilingual Wikipedia contexts shift the locus of bias to the retrieved passages, where cross-lingual claims often conflict. Focusing on East Asian disputes, Kim and Kim (2025) further distinguished between intrinsic training-set model bias from prompt-language inference bias. Extending beyond human consensus, Eibl et al. (2026) audited the LLM-generated *Grokipedia*, finding increasing semantic divergence from Wikipedia across article sections and a greater tendency to position right-leaning content near the top of controversial articles. Creanga et al. (2026) additionally compare prompting strategies for generating summaries that attribute, rather than resolve, conflicting claims, evaluating them with an LLM judge. Finally, historical revisionism audits (Chen et al., 2026) show

that LLMs are largely factual under neutral prompting but readily shift to revisionist narratives when explicitly requested through omission, sanitization, and false balance.

1.4 Research Gap and Project Positioning

Prior work thus characterizes disputed narratives from three complementary angles: edit-history signals showing where editors disagree (Yasseri et al., 2012; Borra et al., 2015; Bykau et al., 2015), corpus-level measures of framing, stance, and sourcing across editions (Oeberst et al., 2020; Xia and Gomez-Zara, 2026; Creanga et al., 2026), and benchmarks of LLM geopolitical bias built largely on short or synthetic prompts (Li et al., 2024a; Kim and Kim, 2025). What remains under-examined is the content of the divergence itself: which specific assertions one edition makes that another omits, contradicts, or attributes differently. Creanga et al. (2026) come closest, extracting and matching conflicting claims within a pipeline for generating attributed summaries. However, their analysis of divergence rests on citation stance and LLM-assigned narrative scores, they report no claim-level breakdown of divergence types, and they do not relate divergence to editorial controversy.

Our project addresses this gap by combining (i) atomic claim extraction followed by cross-lingual claim alignment with compact open-weight language models; (ii) a typology separating omission, contradiction, and differing attribution or framing; and (iii) article- and section-level edit-conflict indicators.

2 Methodology

Our aim is to characterise *what* diverges when two language editions describe the same historically disputed event, and to relate that divergence to editorial conflict. Given two articles A and B on the same event written in different languages, the pipeline aligns their sections (§2.1), decomposes each aligned section into atomic propositions grounded in Wikidata (§2.2) and computes four section-level metrics (§2.3), which are then evaluated with three LLMs (§2.4) and analysed jointly (§2.5). Figure 1 summarises the architecture.

2.1 Cross-Lingual Section Alignment

The sections of both articles are extracted independently, producing $\{S_A^i\}_{i=1}^m$ and $\{S_B^j\}_{j=1}^n$, with named entities linked to Wikidata (Vrandečić and Krötzsch, 2014), this means *Argentina* is annotated with its language-independent identifier `Q414`. These QIDs provide a common semantic reference across language editions and are reused by every subsequent stage.

Because language editions organise the same event under different headings, sections cannot be matched by title. Instead, all candidate section pairs are scored with BERTScore (Zhang et al., 2020) computed from a multilingual encoder, and each section of A is aligned to its most similar counterpart in B:

$$\sigma(i) = \arg\max_j \ \mathrm{BERTScore}\big(S_A^i, S_B^j\big). \quad (1)$$

The resulting aligned pairs $\big(S_A^i, S_B^{\sigma(i)}\big)$ localise divergence within the narrative; for readability we write S_B^j for $S_B^{\sigma(i)}$ below.

2.2 Atomic Propositions and Knowledge Graphs

Each aligned section is decomposed into *atomic propositions* (APs), that is, minimal factual statements that can be evaluated independently. APs are produced with MPROPOSITIONNEUR-V2 (Pommeret et al., 2026) and then passed to QWEN3-8B, which extracts structured triplets $t = (q_s, r, q_o)$, where q_s and q_o are Wikidata QIDs and r is a relation, e.g. `(Q9648, located_in, Q97)`. The triplets of a section form a knowledge graph where nodes are entities and edges are semantic relations.

In parallel, a JSON provenance structure links every triplet back to its source AP, section, article and language, so that candidate matches found in the graphs can be traced back to the exact propositions from which they originated.

2.3 Section-Level Metrics

Four metrics are computed for every aligned section pair. Controversy is an *intra-article* signal derived from edit histories; omission, contradiction and framing are *cross-article* measures of divergence.

2.3.1 Controversy

Controversy is computed independently for each section of each article from the editing behaviour of the wikilinks it contains, following wikilink- and revision-level controversy measures (Borra et al., 2015; Bykau et al., 2015). For each wikilink q in a section we retrieve the revision history of the corresponding article and track the additions, removals,

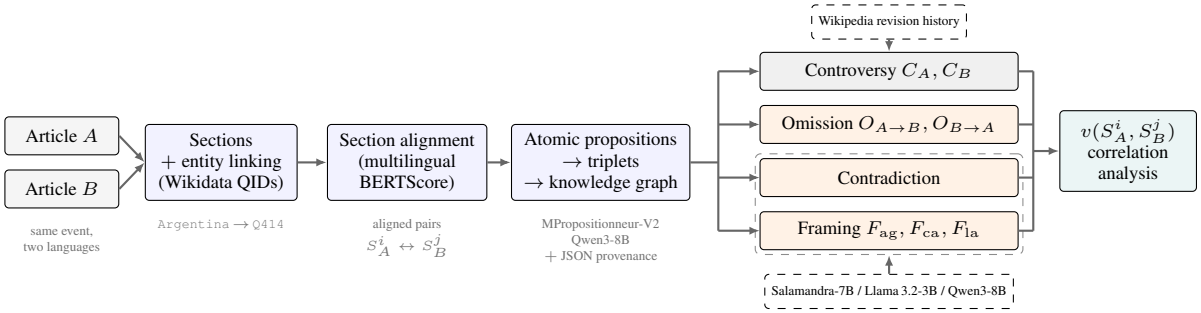


Figure 1: Overview of the proposed pipeline: sections of two articles on the same event are aligned with multilingual BERTScore, decomposed into atomic propositions and normalised into Wikidata-grounded triplets, over which four section-level metrics are computed.

restorations and repeated opposing modifications of q in order to identify conflicting edits. We specifically use the controversy measure developed by Borra et al. (Borra et al., 2015). Let $R_1, \ldots, R_N$ be the revisions considered and $D(R_n)$ the set of sentences in R_n affected by a substantive disagreeing edit. The controversy of q is

$$C(q) \;=\; \sum_{n=1}^{N} \sum_{\substack{x \in D(R_n) \\ q \in x}} \frac{1}{w(x)}, \qquad (2)$$

where $w(x)$ is the number of wiki-linked entities in x, so that the disagreement observed in a sentence is distributed over the entities that may have caused it.

Denoting by Q_A^i the set of wikilinks of section S_A^i, the section-level score is the mean over its wikilinks,

$$C_A\big(S_A^i\big) \;=\; \frac{1}{|Q_A^i|} \sum_{q \in Q_A^i} C_A(q), \qquad (3)$$

with $C_B(S_B^j)$ obtained analogously, so that every aligned pair carries two independent controversy values, one per edition.

2.3.2 Omission

Omission measures whether entities discussed in one section are absent from its aligned counterpart. As it is directional, two scores are computed:

$$O_{A\to B} \;=\; 1 - \frac{\big|Q_A^i \cap Q_B^j\big|}{|Q_A^i|}, \qquad (4)$$

and symmetrically $O_{B\to A}$, with $|Q_B^j|$ in the denominator: $O_{A\to B}$ is the proportion of entities present in A that do not appear in B.

2.3.3 Contradiction

Contradiction is evaluated over atomic propositions rather than isolated entities. The knowledge graphs act as a candidate-retrieval mechanism: triplets from A and B sharing compatible subject and object QIDs, i.e. (q_1, r_A, q_2) and (q_1, r_B, q_2), are taken as potentially corresponding claims. Using the provenance structure, each candidate pair is traced back to its source propositions $AP_A^k \leftrightarrow AP_B^l$, which are submitted to an LLM that decides whether the two statements are contradictory or not. The section-level score is the proportion of contradictory pairs,

$$Contr \;=\; \frac{N_{\text{contradictory}}}{N_{\text{pairs}}}. \qquad (5)$$

2.3.4 Framing

Proposition pairs that refer to equivalent content but are *not* classified as contradictions are analysed for framing differences, i.e. cases where both editions describe the same event but present it differently. Each pair $AP_A^k \leftrightarrow AP_B^l$ is scored by an LLM along three dimensions: **agency** (who is portrayed as acting, initiating or being responsible), **causality** (how causes, consequences and motivations are attributed) and **language** (lexical choice, tone, intensity and sentiment). Each dimension receives an ordinal label of SAME OR EQUIVALENT, SLIGHTLY DIFFERENT, or VERY DIFFERENT, encoded as 0, 1, and 2. Averaging over the N pairs of a section yields a section-level framing profile,

$$F_d \;=\; \frac{1}{N} \sum_{k=1}^{N} f_d^{(k)}, \qquad d \in \{\mathrm{ag}, \mathrm{ca}, \mathrm{la}\}, \qquad (6)$$

where $f_d^{(k)}$ is the encoded score of pair k on dimension d.

2.4 Multi-Model and Human Evaluation

The contradiction and framing metrics rest on LLM judgements and are therefore sensitive to the linguistic and training biases of the model producing them. To mitigate this impact on the evaluation, both analyses are run with three small open-weight models covering a multilingual Spanish-centred, a multilingual English-centred and a third multilingual configuration for tie-breaking: SALAMANDRA-7B (Gonzalez-Agirre et al., 2025), developed at the Barcelona Supercomputing Center, LLAMA 3.2-3B (Grattafiori et al., 2024) and QWEN3-8B (Yang et al., 2025).

We compare the three outputs and report inter-model agreement; pairs on which the models disagree are flagged as uncertain and analysed separately. To assess the reliability of these judgements, a sample of 100 proposition pairs is also annotated by the members of the team, with inter-annotator and model–human agreement reported for contradiction and for each framing dimension. This is particularly important for framing, where the distinction between a semantic difference and a genuine perspectival one is more subjective than entity matching or graph comparison.

2.5 Global Analysis

For every aligned section pair the pipeline outputs the vector

$$v(S_A^i, S_B^j) = \big(C_A,\ C_B,\ O_{A\to B},\ O_{B\to A},\ Contr,\ F_{\text{ag}},\ F_{\text{ca}},\ F_{\text{la}}\big). \quad (7)$$

The final analysis examines the correlation of the dimensions of the vector, testing the hypothesis that higher editorial controversy in one edition is associated with higher omission, contradiction and framing scores in the corresponding section, and thus whether historically disputed topics systematically produce different representations across linguistic editions.

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